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Homing strategies for asymmetric sellers on differentiated platforms

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ABSTRACT

With the flourishing development of e-marketplace platforms, sellers' homing strategies have attracted considerable attention and require careful evaluation of the market environment, especially considering platform competition. This study explores two asymmetric sellers' homing and pricing strategies on two asymmetric platforms. The platforms and sellers are differentiated in their credits and boast varied consumers' willingness to pay. We identify the competition-mitigating effects of utilizing an inferior platform and adopting single-homing strategies. We indicate that such effects should be carefully considered in balance with the valuation-enhancing effect of choosing a superior platform and the market-expanding effect of adopting multi-homing strategy. Moreover, in some situations, the benefits of seller prioritization of platform occupancy and implementation of an "either-or" policy (i.e., mandatory single-homing policy for sellers) can be less than expected. These results enhance our understanding of homing strategy optimization in the face of competition between sellers and platforms.

1. Introduction

The platform economy has been developing rapidly in the past decade. The successes of e-marketplace platforms such as Amazon Marketplace in the U.S. and Tmall in China have led more companies to develop their own platform businesses. For example, Walmart launched its e-marketplace in 2009, which has overtaken Ebay as the second-largest e-commerce player in the U.S., only behind Amazon [30]. In China, when everyone thought the e-commerce industry had settled with two giants, Alibaba and JD.Com, Pinduoduo grow rapidly from an e-commerce upstart into a beast, rivalling Alibaba and JD in terms of active buyers. In 2021, Pinduoduo reached a milestone with annual net profits of 7.8 billion yuan, a turnaround from the 7.2 billion yuan loss in 2020 [32].

The entry of emerging e-commerce platforms has greatly intensified market competition: platforms compete not only for consumers but also for sellers. As a result, sellers can use multiple platforms as selling channels, a behaviour known as multi-homing. For example, with the rise of the Walmart e-marketplace, 7 % of existing Amazon sellers also list products on the Walmart e-marketplace, and another 39 % are considering joining it in 2021 [5]. It has also been reported that 61 % of Amazon sellers sell on at least one other channel in 2022, up from 58 % the year prior, and more than half plan to explore new ecommerce platforms in 2023 [23].

For sellers, the homing strategy requires a careful evaluation of multiple elements. One is the platform's reputation. In China, the main shortcoming of Pinduoduo, compared to Tmall and JD, is its poor public image, which mainly stems from the overrun of counterfeit and shoddy goods in its early years [40]. This has caused consumers to have serious doubts and has lowered their valuations for all sellers on the platform due to so-called reputation externalities, affecting sellers' decision-making on whether to join [33]. For example, Apple has not included Pinduoduo on its authorized online selling channel list, which includes JD and Tmall, owing to concerns about Pinduoduo's reputation as a fake seller. The seller's attributions are another factor to be considered when making a platform-homing decision. The targeting of consumers at low or high market ends, in other words, the "quality" of the selling product, significantly impacts a consumer's purchasing willingness and, thus, should be measured by the sellers. For instance, Xiaomi, which mainly focuses on the mid-to-low-end market in China, opened its official flagship store at Pinduoduo in June of 2018, a strategy in contrast to Apple's. The competition between sellers also complicates

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the problem, as it is necessary to consider competitors' response strategies and their impact when making platform and pricing decisions. Additionally, the intensity of the competition will also affect profits.

Furthermore, the seller's homing strategy influences the platform's interests. For example, to avoid user attrition and transaction volume decline, some platforms adopt an "either-or" policy for their sellers, in which merchants must choose only one of two or more platforms. Merchants who are unwilling to comply with this policy will be refused cooperation or be treated differently by the platform. This phenomenon is common in China's e-commerce platforms and has received serious questioning and considerable attention. It is generally believed that by forcing sellers to take sides and forego sales on other platforms, 'either-or' behaviour directly harms the trading opportunities and economic interests of other platforms and sellers and affects consumers' opportunities and the right to choose. In October 2021, the State Administration for Market Regulation imposed a fine of 3.442 billion yuan on Meituan for forcing merchants to choose between platforms and ordered it to refund merchants 1.289 billion yuan in exclusive cooperation deposits. In December 2023, the Beijing court ruled on JD's lawsuit against the Alibaba Group and its subsidiary, Tmall, determining that Alibaba's "either-or" policy constituted an abuse of market dominance and ordered compensation of 1 billion yuan to JD.

Considering the above issues, this research focuses on sellers' homing strategies for e-marketplaces and their influencing factors. We consider that there are differences in credits between platforms and between the sellers on them. As mentioned above, sellers can differ in terms of brand image, product quality, and service level, while platforms may differ in reputation, regulatory systems, and service capabilities. These differences can affect consumers' purchasing channels and product choices. We seek to address the following questions: How should sellers with different credits choose their platform-homing strategies under competition? What is the value of an inferior platform, and when should it be chosen? When will a single-homing strategy be better than a multi-homing strategy? Is there a first-mover advantage for the seller to enter a platform first? How do these different combinations of homing strategies influence platforms and consumers? From the platform's perspective, will promoting the "either-or" policy always be beneficial? In addition, will prohibiting this policy truly enhance the benefits for sellers and consumers?

To address the above questions, we establish an analytical framework of two sellers, two platforms, and a mass of consumers with heterogeneous valuations for a product. The credit differentiation between sellers and platforms is characterized by a stylized willingness-to-pay model. On the sellers' side, we develop a two-stage game in which the sellers first choose their platform-homing strategies and then determine selling prices on the selected platform(s). By analysing the pricing and homing strategies within this modelling framework, we obtain the following results. First, an inferior platform should be used as a cushion for competition because the benefits brought about by softening the competition may surpass the decrease in product value caused by using an inferior platform. For example, when the credit difference between sellers is in the middle range, low-credit sellers should abandon the superior platform and choose the inferior one. Second, in a competitive environment, single-homing is not a subtype of multi-homing but rather a strategy that may be superior to multisourcing, as the competition-intensifying effect of multi-homing can outweigh its market-expanding effect and, thus, become disadvantageous to sellers. For example, in a situation where the difference in reputation between sellers is small, both high- and low-credit sellers will adopt a single-homing strategy. Third, the advantage of being a first mover only exists when the credit difference between platforms is small or when the credit differences between platforms and sellers are moderate; in other cases, being the first to enter a platform does not confer an advantage. Finally, the "either-or" policy is not only disadvantageous to the inferior platform but may also cause losses to the superior platform, creating a lose-lose situation. On the other hand, this policy can be beneficial for both low-credit sellers and consumers under some conditions, contradicting conventional wisdom that banning this policy will benefit both sellers and consumers. These results have not been found in previous studies and fill the research gap on homing strategies of asymmetric sellers on differentiated platforms.

The remainder of this paper is organized as follows. In Section 2, we review the literature related to our work. Section 3 describes the model with notations and assumptions that are used in the following analysis. In Section 4, we develop the sellers' optimal pricing decisions and the profits of the involved parties under different strategy combinations. In Section 5, we discuss the sellers' equilibrium homing strategies and their implications. In Section 6, we extend our model and show the robustness of the main results. Finally, we conclude the paper with key findings and future research directions in Section 7. All relevant proofs can be found in the Appendix.

2. Literature review

Our research is related to two streams of literature. One is product competition among heterogeneous consumers, and the other deals with the multi-homing behaviours of sellers.

Regarding the first stream, extensive studies have focused on the heterogeneous preferences of consumers for product quality. Chambers et al. [8] study competitive behaviour in a duopoly where two manufacturers compete on quality and price. Li and Chen [27] further examine a common retailer downstream of the supply chain and discuss the impact of two manufacturers' different pricing timing choices. Geng et al. [15] demonstrate the impact of social interactions on firms' quality differentiation and profit performance in a duopoly competition framework. Other literature incorporates factors such as personalized pricing [13], channel encroachment ([16,41]; Zhao et al. [47]) and quality disclosure [6,42]. In addition, there have been papers on the heterogeneous preferences of consumers for different channels. Li et al. [25] consider that a product sold through a retailer is preferred by consumers when the product is sold through both the retailer channel and the online direct channel. They study the introduction strategy of a store brand via an online channel in a supply chain with one retailer and one manufacturer. Zhang et al. [43] assume that consumers have less preference for a direct channel and examine manufacturer encroachment and retailer product introduction strategies under three kinds of products: the manufacturer's product, the retailer's lower-quality product and the retailer's higher-quality product. Wan et al. [37] also consider that a manufacturer can encroach on the retail market by introducing a direct channel that is less accepted by consumers than a traditional channel and show the impact of retail pricing leadership on manufacturers' encroachment decisions. Chang et al. [9] further assume that the manufacturer can provide vertically differentiated products and determine the encroachment strategy into retail market. In contrast to the above papers, our study considers consumers' heterogeneous preferences for both platforms and products (i.e., sellers).

In the second literature stream on platform competition, most research is related to the pricing strategies of competitive two-sided platforms [1–4,17,29,35,39]. In addition to pricing optimization, other platform operational management strategies studied include bundling [12,28], information revelation [10,18], platform investment [48], wage schemes [19,44], and quality regulation [20,31]. In this regard, our paper is more related to the endogenous homing decision problem, on which there is limited research. Karle et al. [24] show the influence of competition between two symmetric sellers on the platform market structure. They find that tough competition induces sellers to join different platforms and that soft competition leads to agglomeration on a single platform. Jeitschko and Tremblay [21] build a two-sided market model in which both sides of users endogenously decide their homing strategy and study the interactions between platforms' pricing strategies and users' endogenous homing. Following this paper, Pu et al. [34] consider a supply chain consisting of one manufacturer and two

platforms and study the manufacturer's optimal homing policy and the platform's optimal selling format. Dai et al. [14] assume a manufacturer already has an online reselling channel and investigate the incentive for channel expansion under various strategies: agency selling in the incumbent platform, agency selling, or reselling in the asymmetric competing platform. They find that the manufacturer's channel expansion strategies depend on the market potential ratio, competition intensity, and agency channel costs. In contrast to these studies, our paper considers the homing strategies of two sellers with vertical differentiation, i.e., one seller sells a product of higher credit than the other through differentiated platforms.

In summary, our paper contributes to the literature by examining the homing and pricing strategies for asymmetric sellers on differentiated platforms. We consider the heterogeneous preferences of consumers for both the sellers' products and the platforms themselves. Our findings emphasize the potential benefit of using a less competitive platform to reduce competition between sellers. We also reveal that adhering to one platform can be more advantageous than using multiple platforms if the competition-mitigating effect of the former outweighs the market-expanding effect of the latter. Additionally, we investigate the circumstances in which suppliers may have a first-mover advantage and the feasibility of a platform's "either-or" policy. The results enrich the research landscape on strategy optimization in the face of competition between sellers and platforms. Table 1 presents a list of the studies most closely related to our paper and compares them with ours.

3. Model description

We consider two platforms, A and B, and two sellers, H and L. Platform A has a higher credit than Platform B for consumers since it has a better reputation, credibility, service capacity, etc. That is, consumers prefer to buy a product on Platform A over Platform B, all else being equal. Specifically, we use $\alpha \in (0,1)$ to denote the relative credit level of the inferior platform, i.e., if a consumer's willingness to pay for a product sold on Platform A is 1, then for the same product sold on Platform B, the willingness to pay is α. In addition, Seller H has more credit than Seller L due to a sounder brand, higher product quality, better service level, etc. We use $\beta \in (0,1)$ to denote the relative credit level of the inferior seller, i.e., if a consumer's willingness to pay for a product from Seller H is 1, then that from Seller L is β. Each seller chooses the platform(s) with three possible strategies: selling only on Platform A (strategy A), selling only on Platform B (strategy B), and selling on both platforms (strategy AB). We use (m, n) to denote the homing strategy combination for the two platforms, in which m and n represent the strategies of Sellers H and L, respectively, with $m, n \in \{A, B, AB\}$.

In addition, each seller determines the selling price(s) on the chosen platform(s). Denote p_{ij} as the selling price posed by Seller i on Platform j (if the platform is chosen), where $i = H, L$ and $j = A, B$. Hence, a customer with marginal quality valuation θ will obtain a net utility: (a) $u_{HA} = \theta - p_{HA}$ if he or she purchases from Seller H on Platform A; (b) $u_{HB} = \alpha\theta - p_{HB}$ if he or she purchases from Seller H on Platform B; (c) $u_{LA} = \beta\theta - p_{LA}$ if he or she purchases from Seller L on Platform A; and (d) $u_{LB} = \beta\alpha\theta - p_{LB}$ if he or she purchases from Seller L on Platform B. Note that this paper posits that the platform's credit is related to consumers' perceptions of sellers' credit. For instance, a larger, more reputable and trustworthy e-commerce platform will endorse the merchants that operate on it, thereby positively influencing consumers' perceptions of their products [11,36,45]. Consequently, the two utility discount factors can be intertwined and represented in a multiplicative relationship, which has been employed by the existing studies, e.g., in Zhang et al. [43] and Wei and Xu [38]. We draw upon this type of modelling approach to characterize the consumer's willingness-to-pay (Section 6.3 discusses an alternative model of two-dimensional differentiation and shows the main results carry over). Assuming that θ is heterogeneous among consumers and uniformly distributed on $[0,1]$ [16,46], we can formulate the demand function of Seller i on Platform j under strategy combination (m, n), denoted by $D_{ij}^{(m,n)}$ and illustrated by Table A1 in the Appendix.

The commission rate for Platform j is denoted as $\lambda_j \in (0,1)$ and is considered a fixed parameter [7,22,26]. For ease of computation and presentation, we assume that the two platforms' commission rates are equal, i.e., $\lambda_A = \lambda_B = \lambda$. We will examine the scenario of different commission rates in Section 6.1 and the scenario of endogenous commission rates in Section 6.2, both of which demonstrate that the core results remain unchanged. We also assume zero production costs for both sellers. Therefore, the profit functions of the sellers and platforms under strategy combination (m, n) are given by

$$\begin{cases} \pi_H^{(m,n)} = (1-\lambda)(p_{HA}^{(m,n)} D_{HA}^{(m,n)} + p_{HB}^{(m,n)} D_{HB}^{(m,n)}) \\ \pi_L^{(m,n)} = (1-\lambda)(p_{LA}^{(m,n)} D_{LA}^{(m,n)} + p_{LB}^{(m,n)} D_{LB}^{(m,n)}) \end{cases} \text{and}$$

$$\begin{cases} \pi_A^{(m,n)} = \lambda(p_{HA}^{(m,n)} D_{HA}^{(m,n)} + p_{LA}^{(m,n)} D_{LA}^{(m,n)}) \\ \pi_B^{(m,n)} = \lambda(p_{HB}^{(m,n)} D_{HB}^{(m,n)} + p_{LB}^{(m,n)} D_{LB}^{(m,n)}), \end{cases}$$

Table 2
Notations and descriptions.

Notations	Descriptions
α	Credit level of Platform B (relative to that in Platform A)
β	Credit level of Seller L (relative to that of Seller H)
θ	Consumer's valuation for unit product credit
u_{ij}	Net utility of a consumer when purchasing from Seller $i = H, L$ on Platform $j = A, B$
λ_j	Commission rate for Platform $j = A, B$
λ	Commission rate when $\lambda_A = \lambda_B$
$p_{ij}^{(m,n)}$	Selling price of Seller $i = H, L$ on Platform $j = A, B$ under strategy combination (m, n) in which $m, n = A, B$ or AB
$D_{ij}^{(m,n)}$	Demand of Seller $i = H, L$ on Platform $j = A, B$ under strategy combination (m, n) in which $m, n = A, B$ or AB
$\pi_i^{(m,n)}$	Profit of Seller $i = H, L$ under strategy combination (m, n) in which $m, n = A, B$ or AB
$\pi_j^{(m,n)}$	Profit of Platform $j = A, B$ under strategy combination (m, n) in which $m, n = A, B$ or AB
$U^{(m,n)}$	Total consumer surplus under strategy combination (m, n) in which $m, n = A, B$ or AB

Table 1
Model comparison between this paper and related research.

	Competition between sellers	Competition between platforms	Endogenous homing strategy	Asymmetric sellers/ products	Asymmetric platforms
Li and Chen [27]	√	x	x	√	x
Bakos and Halaburda [2]	x	√	x	x	x
Jeitschko and Tremblay [21]	x	√	√	x	√ (in extension)
Karle et al. [24]	√	√	√	x	x
Geng et al. [15]	√	x	x	√	x
Pu et al. [34]	x	√	√	x	x
Dai et al. [14]	x	√	√	x	√
Our paper	√	√	√	√	√

where $D_{HB}^{(A,n)} = 0$, $D_{HA}^{(B,n)} = 0$, $D_{LB}^{(m,A)} = 0$, and $D_{LA}^{(m,B)} = 0$.

The sequence of events is as follows. First, the two sellers simultaneously or sequentially choose their homing strategies; then, the sellers simultaneously set their prices; and finally, consumers make purchase decisions. For convenience, we refer to Seller H as "she'', Seller L as "he", and both platforms as "it". Table 2 lists the notations used in this paper.

4. Price and profit analysis

4.1. Single-homing cases

In this subsection, we analyse the equilibrium selling prices and the profits of the sellers and platforms when both sellers adopt a single-homing strategy, i.e., $m, n = A$ or B, in the homing strategy combination (m, n). Through backward induction, we obtain the equilibrium outcomes presented in Table A2 in the Appendix. Then, we compare the sellers' selling prices under single-homing strategies.

Proposition 1. (a) *For Seller H, we have* $p_{HA}^{(A,B)*} > \max\left\{p_{HA}^{(A,A)*}, p_{HB}^{(B,A)*}, p_{HB}^{(B,B)*}\right\}$;

(b) *For Seller* L, *we have* $\min\left\{p_{LA}^{(A,A)*}, p_{LA}^{(B,A)*}, p_{LB}^{(A,B)*}\right\} > p_{LB}^{(B,B)*}$. *Moreover,* $p_{LB}^{(A,B)*} > \max\left\{p_{LA}^{(A,A)*}, p_{LA}^{(B,A)*}\right\}$ *holds if and only if* $\alpha > 0.57$ *and* $\beta_1 < \beta < \beta_2$.

Note: the expressions of β_1 *and* β_2 *are given in the Appendix.*

From Proposition 1, we can see that Seller H will set the highest selling price in strategy combination (A, B), where seller competition is the weakest among all strategy combinations. On the other hand, Seller L will set the lowest selling price in strategy combination (B, B) in response to the fierce competition with Seller H. Moreover, the selling price for Seller L is the highest in strategy combination (A, B) if his credit is in a medium range and the credit of Platform B is sufficiently high. This is an interesting result showing that Seller L selling on an inferior platform can charge a higher price than selling on a superior platform. The reason for this is that joining Platform B rather than Platform A has two effects for Seller L: it decreases the product valuation perceived by consumers, thereby reducing their willingness to pay, which leads to a lower selling price, and it softens the competition with Seller H, which motivates a higher selling price. We refer to the former as the valuation-reducing effect and the latter as the competition-mitigating effect. When the credit of Seller L is in a medium range and that of Platform B is sufficiently high, the competition-mitigating effect prevails over the valuation-reducing effect, allowing Seller L to set a higher price on the inferior platform.

Proposition 2. *When both sellers adopt a single-homing strategy, we have:*

(a) $\pi_H^{(A,B)*} > \pi_H^{(B,B)*}$ *always holds;* $\pi_H^{(B,A)*} > \pi_H^{(A,A)*}$ *holds if and only if* $\alpha_1 < \alpha < \alpha_2$ *and* $\beta > 0.95$*;*

(b) $\pi_L^{(A,B)*} > \pi_L^{(A,A)*}$*holds if and only if* $\beta > \beta_3$*;* $\pi_L^{(B,B)*} > \pi_L^{(B,A)*}$ *holds if and only if*$4\beta(4-\beta)^2(-\alpha+\beta) < (1-\beta)(\alpha-4\beta)^2\alpha$ *and* $\beta > \alpha\beta_3$.

Note: the expressions of α_1, α_2 *and* β_3 *are given in the Appendix.*

Proposition 2 indicates that selling on a higher credit platform is not always beneficial to a seller. If Seller L has chosen the inferior platform, i.e., Platform B, it is straightforward that Seller H would like to choose Platform A to enjoy a higher platform credit and avoid competition. On the other hand, given that Seller L has chosen Platform A, it is preferable for Seller H to join Platform B when the credit difference between the sellers is minimal and that between the platforms is in a medium range. In this case, the competition between the two sellers is radical since their credits are very close. Choosing Platform B allows Seller H to avoid intense competition with Seller L on the same platform, resulting in the competition-mitigating effect. On the other hand, selling on Platform B diminishes the perceived credibility of the product compared to Platform A, referred to as the valuation-reducing effect. For Seller H, the benefit of the competition-mitigating effect outweighs the disadvantage of the valuation-reducing effect, under the condition that the credit difference between the platforms is moderate: if Platform B has a very high credit, the competition-mitigating effect becomes limited; if Platform B has a very low credit, the loss from the valuation-reducing effect becomes significant. Only when the credit of Platform B falls within a moderate range can Seller H effectively leverage it to alleviate fierce competition and achieve greater profit than by joining Platform A.

Proposition 2 also prescribes the optimal choice for Seller L. Similarly, a small difference between the sellers can cause intense competition and compel Seller L to choose the lower credit platform to soften their competition. Specifically, Seller L's preference, given that Seller H has chosen Platform B, is illustrated in Fig. 1 (we assume $\lambda = 0.1$ for illustrative purposes). In this case, it is interesting to note that Seller L would rather join Platform B together with Seller H if the platforms' credit difference is within a medium range. The rationale is similar to that above, that is, the softened competition with Seller H generates a greater profit for Seller L than increasing the consumer's willingness to pay by joining Platform A. This again highlights the value of the inferior platform, which can be used as a cushion to soften competition when the difference between sellers is small.

4.2. Multi-homing cases

In this subsection, we first consider that Seller H adopts a multi-homing strategy, while Seller L adopts a single-homing strategy, i.e., $m = AB$ and $n = A$ or B in the homing strategy combination (m, n). In this case, Seller H determines two selling prices, p_{HA} and p_{HB}, on the two platforms, while Seller L sets the selling price p_{Ln} on his chosen platform for either $n = A$ or B. By applying backward induction, the equilibrium outcomes are summarized in Table A3 in the Appendix.

Then, we suppose that Seller L adopts a multi-homing strategy and that Seller H may sell on Platform A, Platform B or both, i.e., $m = A$, B or AB and $n = AB$, in the homing strategy combination (m, n). Specifically, we make the following observations: (i) the strategy combination (A, AB) reduces to (A, A); (ii) when $\alpha > \beta$, the strategy combination (B, AB) reduces to (B, A); (iii) and when $\alpha > \beta$, the strategy combination (AB, AB) reduces to (AB, A). Note that adding Platform B as

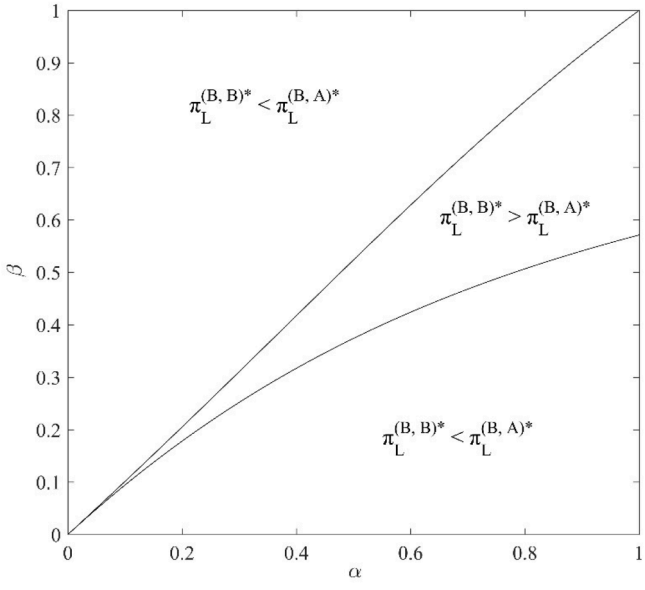


Fig. 1. Comparison of Seller L's profits under strategy combinations (B, B) and (B, A) ($\lambda = 0.1$).

an additional selling channel can divert Seller L's original demand from Platform A, resulting in a loss due to diminished marginal profit from sales on Platform B. This is usually referred to as channel cannibalization. In this case, if Seller H sells on Platform A ($m = A$) or if the credit difference between platforms is greater than that between sellers ($\alpha > \beta$), Seller L chooses to set a sufficiently high price on Platform B, effectively reducing its demand on that platform to zero and maintaining the market share on Platform A. By this way, channel cannibalization is eliminated and multi-homing is transformed into single-homing. Excluding these three multi-homing cases, we provide the prices and profits for $m = B$orAB and $n = AB$ when $\alpha < \beta$ in Table A4 in Appendix.

Proposition 3. *We have (a) $p_{HA}^{(A,AB)*} > p_{HB}^{(B,AB)*}$; (b) $p_{LB}^{(AB,B)*} > p_{LA}^{(AB,A)*}$ holds if and only if $\beta_4 < \beta < \beta_5$.*

Note: the expressions of β_4 and β_5 are given in the Appendix.

Proposition 3 indicates that Seller H will always set a higher price on Platform A than on Platform B when her competitor is multi-homing. This is because the consumer's willingness to pay for the product on Platform A is greater and the competition is weaker in strategy combination (A, AB) than in (B, AB). Proposition 3 also shows that Seller L will set a higher price on Platform B if his credit level is in the medium range, in which case competition-mitigating effect outweighs the valuation-reducing effect in choosing an inferior platform.

Proposition 4. *We have (a)$\pi_H^{(A,AB)*} > \pi_H^{(B,AB)*}$; (b) $\pi_L^{(AB,B)*} > \pi_L^{(AB,A)*}$ holds if and only if $\beta > \alpha\beta_3$ and$\alpha\left((2\beta+1)\alpha+\beta^2-4\beta\right)^2 > \beta(\beta-\alpha)(1-\alpha)(4-\beta)^2$.*

Proposition 4 shows that Seller H gains more profit by joining Platform A when Seller L adopts the multi-homing strategy, which is an intuitive result. In contrast, given that Seller H chooses the multi-homing strategy, Seller L prefers to join Platform B when the credit difference between sellers is within a medium range, as depicted in Fig. 2. As explained in Proposition 2, in this case, mitigated competition benefits Seller L more than a higher credit platform.

4.3. Impact of multi-homing behaviour

By comparing the single-homing and multi-homing cases, we can obtain the impacts of multi-homing behaviour on supply chain members. We first present the results on the seller's pricing decisions.

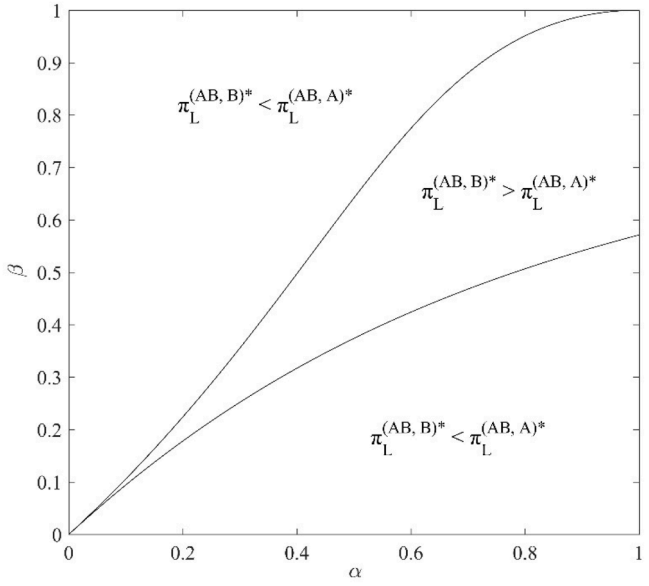


Fig. 2. Comparison of Seller L's profits under strategy combinations (AB, B) and (AB, A) ($\lambda = 0.1$).

Proposition 5. *On the selling prices, we have:*

(a) $p_{iA}^{(A,A)*} = p_{iA}^{(A,AB)*} > \max\left\{p_{iA}^{(AB,A)*}, p_{iA}^{(AB,AB)*}\right\}$ *for* $i = H, L$;

(b) $p_{HA}^{(A,B)*} > \max\left\{p_{HA}^{(AB,B)*}, p_{HA}^{(A,AB)*}, p_{HA}^{(AB,AB)*}\right\}$, $p_{LB}^{(A,B)*} > \max\left\{p_{LB}^{(AB,B)*}, p_{LB}^{(A,AB)*}, p_{LB}^{(AB,AB)*}\right\}$;

(c) $p_{HB}^{(B,A)*} > \max\left\{p_{HB}^{(AB,A)*}, p_{HB}^{(B,AB)*}, p_{HB}^{(AB,AB)*}\right\}$ *and* $p_{LA}^{(B,A)*} > \max\left\{p_{LA}^{(AB,A)*}, p_{LA}^{(B,AB)*}, p_{LA}^{(AB,AB)*}\right\}$ *hold if* $\alpha < \beta$; $p_{HB}^{(B,A)*} = p_{HB}^{(AB,A)*} = p_{HB}^{(B,AB)*} = p_{HB}^{(AB,AB)*}$ *and* $p_{LA}^{(B,A)*} = p_{LA}^{(AB,A)*} = p_{LA}^{(B,AB)*} = p_{LA}^{(AB,AB)*}$ *hold if* $\alpha > \beta$;

(d) $p_{iB}^{(B,B)*} = p_{iB}^{(AB,B)*} > \max\left\{p_{iB}^{(B,AB)*}, p_{iB}^{(AB,AB)*}\right\}$.

Proposition 5 shows that the multi-homing behaviours of sellers generally reduce their selling prices, or at least remain prices unchanged. The price reduction occurs due to the cannibalization among multiple platforms utilized by the seller and intensified competition with rival sellers, in contrast to single homing. On the other hand, as shown in Propositions 5(a) and (c), the multi-homing behaviour of Seller L on Platform A does not influence the selling prices when both sellers sell on Platform A or when Seller H sells on Platform B and Seller L sells on Platform A, with $\alpha > \beta$. As explained in Section 4.2, these two multi-homing cases reduce to the single-homing cases, and thus, prices remain unchanged. Moreover, Propositions 5(c) and (d) indicate that the multi-homing behaviour of Seller H on Platform B does not affect prices when both sellers sell on Platform B or when Seller H sells on Platform B and Seller L sells on Platform A, with $\alpha > \beta$. In these two cases, Seller H entering another marketplace, Platform A, does not influence her price on Platform B, and Seller L's price remains unchanged.

Corollary 1. *On the sellers' profits, we have:*

(a) *If Seller H shifts from single homing to multi-homing, her profit will decrease in (A, A) and (A, B), increase in (B, B), and may either decrease or increase in (B, A),* i.e. $\pi_H^{(A,A)^*} > \pi_H^{(AB,A)^*}$; $\pi_H^{(A,B)^*} > \pi_H^{(AB,B)^*}$;$\pi_H^{(B,B)^*} < \pi_H^{(AB,B)^*}$; *and* $\pi_H^{(B,A)*} < \pi_H^{(AB,A)*}$ *holds if and only if* $4\alpha\beta\left(\beta^2+(2\alpha-4)\beta+\alpha\right)^2(\alpha-\beta) > T(1-\beta)(\alpha-4\beta)^2$;

(b) *If Seller* L *shifts from single homing to multi-homing, his profit will remain unchanged in (A, A), may either decrease or increase in (A, B) and (B, B), and may either decrease or remain unchanged in (B, A),* i.e. $\pi_L^{(A,A)^*} = \pi_L^{(A,AB)^*}$; $\pi_L^{(A,B)*} > \pi_L^{(A,AB)*}$ *holds if and only if* $\beta > \beta_3$; $\pi_L^{(B,A)^*} > \pi_L^{(B,AB)^*}$ *holds if* $\beta > \alpha$ *and* $\pi_L^{(B,A)^*} = \pi_L^{(B,AB)^*}$ *holds if* $\beta < \alpha$; $\pi_L^{(B,B)^*} > \pi_L^{(B,AB)^*}$ *holds if and only if* $\beta > \alpha\beta_3$ *and*$4\alpha(1-\beta)\left((2\beta+1)\alpha+\beta^2-4\beta\right)^2 > \beta(\beta-\alpha)(4-\beta)^2\left((10\beta-1)\alpha^2+\left(5\beta^2-23\beta\right)\alpha+\beta(4-\beta)^2\right)$.

Corollary 1 shows the impacts of the multi-homing strategy on the sellers' profits. It can be seen that Seller H, if selling on Platform A, will be reluctant to join Platform B. This demonstrates that, in the context of seller competition, multi-homing can be suboptimal compared to single-homing (even if there is no expense such as a fixed cost for selling on more platforms). In fact, it is evident that multi-homing involves utilizing more selling channels than single-homing, thereby benefiting the seller by expanding market reach; we refer to this as the market-expanding effect. On the other hand, introducing additional marketplaces leads to more intense competition and lower selling prices, which can undermine the seller. This phenomenon is referred to as the competition-intensifying effect in this paper. It is shown that the market-expanding effect is suppressed by the competition-intensifying effect for Seller H who sells on Platform A. However, Seller H will prefer joining Platform A in most cases if she is already on Platform B. This underscores the value of a higher credit platform for the seller. More specifically, Seller H on Platform B will reject joining Platform A only if Seller L is

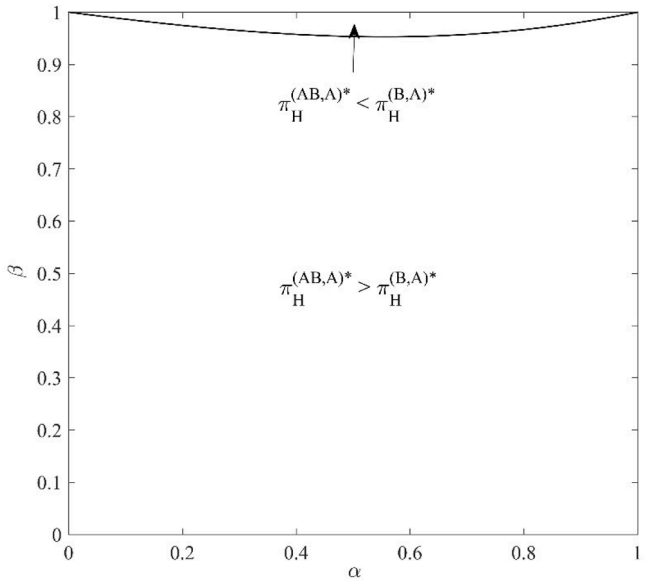


Fig. 3. Comparison of Seller H's profits under strategy combinations (AB, A) and (B, A) ($\lambda = 0.1$).

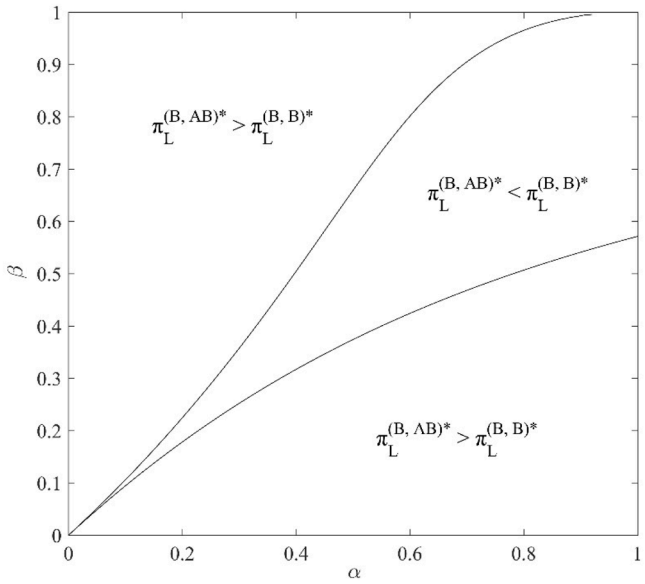


Fig. 4. Comparison of Seller L's profits under strategy combinations (B, B) and (B, AB) ($\lambda = 0.1$).

committed to Platform A and the credit difference between the two sellers is significantly small, as shown in Fig. 3. In this case, the market-expanding effect of joining Platform A is dominated by its competition-intensifying effect. As a result, Seller H prefers to sell on a lower-credit platform rather than engage in multi-homing.

On the other hand, if Seller H has joined Platform A, choosing a multi-homing strategy is equivalent to only joining Platform A for Seller L since the case of (A, AB) reduces to (A, A), as mentioned in Section 4.2. This multi-homing strategy is also less favourable than exclusively joining Platform B, if Seller L has sufficiently high credit. This situation once again exemplifies how the valuation-enhancing effect of selling through a superior platform can be overshadowed by the competition-mitigating effect of opting for an inferior platform. Furthermore, for Seller L, given that he has been on Platform A and his opponent has been on Platform B, reaching out to a superior platform is never an optimal homing policy for him. Specifically, if the relative credit of Platform B is lower than that of Seller L, a multi-homing strategy is detrimental to Seller L since the competition-intensifying effect outweighs the market-expanding effect. In addition, given that both sellers are on Platform B, Seller L would also like to sell on Platform A if and only if the difference between the relative credits of Seller L and Platform B is sufficiently large, as illustrated in Fig. 4, which enables Seller L to benefit more from adding Platform A as another selling channel.

Corollary 2. *We have* $\pi_A^{(AB,A)^*} < \pi_A^{(A,A)^*}$, $\pi_A^{(AB,B)^*} < \pi_A^{(A,B)^*}$, $\pi_B^{(AB,A)^*} < \pi_B^{(B,A)^*}$ *and* $\pi_B^{(AB,B)^*} < \pi_B^{(B,B)^*}$.

Corollary 2 shows that Seller H's multi-homing behaviour is detrimental to the platform to which she originally belonged, given that Seller L's homing strategy is fixed. In fact, the introduction of a high-credit seller to a new platform will induce fiercer competition between sellers and channels, incur lower selling prices and consumer churn, both of which are unfavourable to the original platform. This can explain why some marketplace platforms, such as Tmall, JD and Amazon, used to try to prohibit their merchants, especially renowned brands, from joining another platform.

Corollary 3. *We have* $U^{(A,A)*} < U^{(AB,A)*}$, $U^{(A,B)*} < U^{(AB,B)*}$, $U^{(B,A)*} < U^{(AB,A)*}$ *and* $U^{(B,B)*} < U^{(AB,B)*}$.

Corollary 3 shows that the multi-homing behaviour of Seller H will increase consumer surplus. This is because adding another platform by Seller H will decrease selling prices and stimulate more purchases.

5. Equilibrium homing strategies

This section provides the sellers' equilibrium homing strategies. We consider three decision sequences: (a) Sellers H and L determine the homing strategies simultaneously, (b) Seller L moves first and Seller H follows, and (c) Seller H moves first and Seller L follows.

Proposition 6. *When Sellers H and* L *move simultaneously, (B, A) and (A, B) are two Nash equilibria if* $\alpha_1 < \alpha < \alpha_2$ *and* $\beta > 0.95$, *with* $\pi_H^{(A,B)*} > \pi_H^{(B,A)*}$ *and* $\pi_L^{(A,B)*} < \pi_L^{(B,A)*}$; *(A, A) is the unique Nash equilibrium*[1] *if* $\beta < \beta_3$; *and (A, B) is the unique Nash equilibrium otherwise.*

The result of Proposition 6 is visually presented in Fig. 5. It can be observed that if the difference in credit between the sellers is significantly small, then there exist two pure strategy Nash equilibria: one seller joins Platform A, and the other seller joins Platform B, and vice versa. In this case of intense competition, multi-homing is a dominated strategy for both parties since the competition-intensifying effect overweighs the market-expanding effect. Among the single-homing strategies, a seller on Platform B will be taken advantage of by a seller on Platform A, but they will both lose more if they both join platform A. As the credit difference between sellers increases, joining Platform A becomes the dominant strategy for Seller H, regardless of the choice of Seller L. In this case, if Seller L has relatively high credit, he will choose Platform B; otherwise, he will choose Platform A. It is worth noting that all equilibrium strategies are single-homing. In fact, according to Corollary 1, if Seller H decides to sell on Platform A, she will always resist a multihoming strategy, and Seller L will prefer to sell only on Platform B if his credit is higher than β_1. Corollary 1 also precludes the motivation to adopt a multihoming strategy for Seller L if Seller H has chosen Platform B and the credit difference between the sellers is significantly small.

Proposition 7. *When Seller* L *moves first, (B, A) is the Stackelberg equilibrium if* $\alpha_1 < \alpha < \alpha_2$ *and* $\beta > 0.95$, *with* $\pi_H^{(A,B)*} > \pi_H^{(B,A)*}$ *and*

[1] As shown in Section 4.2, (A,AB) reduces to (A,A) so we do not take (A,AB) as an effective equilibrium and refer to all equilibria as single-homing strategies.

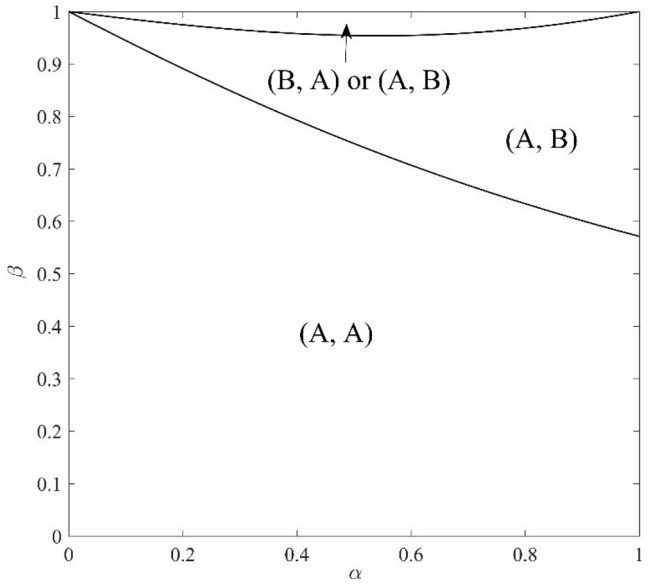


Fig. 5. The equilibrium homing strategies when sellers move simultaneously ($\lambda = 0.1$).

$\pi_L^{(A,B)*} < \pi_L^{(B,A)*}$; *(A, A) is the Stackelberg equilibrium if* $\beta < \beta_3$; *and (A, B) is the Stackelberg equilibrium otherwise.*

Proposition 7 gives the equilibrium homing strategies when Seller L moves first, as illustrated in Fig. 6. The results are the same as when both sellers move simultaneously, with the exception that only (B, A) is an equilibrium if the credit difference between the sellers is significantly small. In this narrow region, Seller L takes the first-mover advantage to join Platform A first and pushes Seller H to Platform B.

Proposition 8. *When Seller H moves first, (A, B) is the Stackelberg equilibrium if* $\beta > \beta_3$; *(AB, B) is the Stackelberg equilibrium if* $\alpha\beta_3 < \beta < \beta_3$ *and* $\beta(1-\alpha)(\beta-\alpha)(4-\beta)^2 < \left((2\beta+1)\alpha+\beta^2-4\beta\right)^2\alpha$; *and (A, A) is the Stackelberg equilibrium otherwise.*

Fig. 7 illustrates the results of Proposition 8 and shows that the Stackelberg equilibrium when Seller H moves first is the same as the Nash equilibrium when both sellers move simultaneously, with two

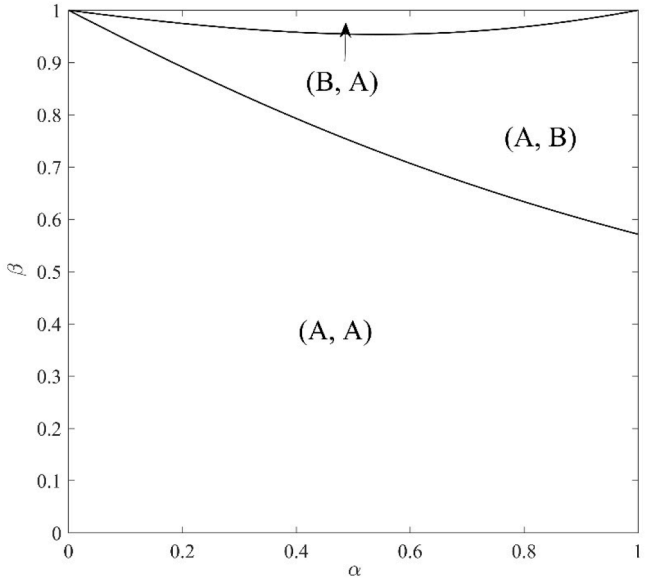


Fig. 6. The equilibrium homing strategies when Seller L moves first ($\lambda = 0.1$).

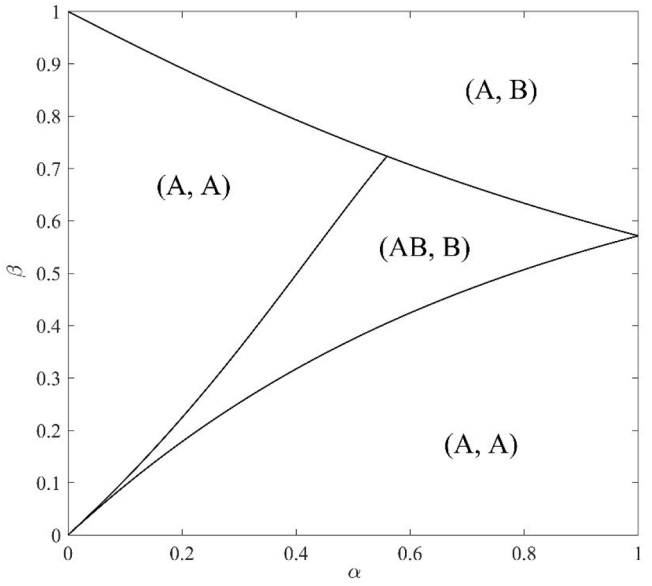


Fig. 7. The equilibrium homing strategies when Seller H moves first ($\lambda = 0.1$).

exceptions. First, (A, B) is the equilibrium if the credit difference between the sellers is significantly small, again showing the first-mover advantage in determining homing strategies. Second, when the credit differences between sellers and platforms are both in the medium range, (AB, B) instead of (A, A) becomes the Stackelberg equilibrium. In this case, as previous analyses indicate, Seller L will opt for the inferior platform, Platform B, to mitigate competition if Seller H either joins only Platform B or participates in both platforms. Conversely, Seller L will select Platform A if Seller H chooses to join Platform A. Anticipating this strategic response, Seller H's optimal strategy is multi-homing, which can deter Seller L from joining Platform A, thereby allowing Seller H to capture a larger market share. Furthermore, the benefit of the market-expanding effect of multi-homing outweighs the drawback of the competition-intensifying effect for Seller H in this case, as demonstrated by Corollary 1.

In the following, we investigate how the moving sequences of sellers impact the platforms and consumers. There are two cases in which equilibrium homing strategies change based on the order of movements: (a) When $\alpha_1 < \alpha < \alpha_2$ and $\beta > 0.95$, the equilibrium is (B, A) if Seller L moves first, and (A, B) if Seller H moves first; (b) When $\alpha\beta_3 < \beta < \beta_3$ and $\beta(1-\alpha)(\beta-\alpha)(4-\beta)^2 < \left((2\beta+1)\alpha+\beta^2-4\beta\right)^2\alpha$, the equilibrium is (AB, B) if Seller H moves first, which differs from (A, A) when Seller L moves first or when both sellers move simultaneously. We first examine the consumer surplus, which is denoted by $U^{(m,n)}$ under strategy combination (m, n).

Corollary 4. *(a) When* $\alpha_1 < \alpha < \alpha_2$ *and* $\beta > 0.95$, *we have* $U^{(A,B)*} < U^{(B,A)*}$; *(b) when* $\alpha\beta_3 < \beta < \beta_3$ *and* $\beta(1-\alpha)(\beta-\alpha)(4-\beta)^2 < \left((2\beta+1)\alpha+\beta^2-4\beta\right)^2\alpha$, *we have* $U^{(AB,B)*} < U^{(A,A)*}$.

Corollary 4 indicates that the consumer surplus is the highest when Seller L moves first and lowest when Seller H moves first (in the regions where equilibrium strategies change with the moving sequences). More specifically, when Seller L moves first, (B, A) instead of (A, B) is the equilibrium if the credit of Seller L is very close to that of Seller H. This increases the utility of purchasing from Seller L and causes a lower price since $p_{LA}^{(B,A)*} < p_{LB}^{(A,B)*}$, as stated in Proposition 1(b). In this case, the consumers benefit from Seller L moving first. On the other hand, when Seller H moves first, the caused equilibrium (AB, B) forces consumers with a low willingness-to-pay to purchase from Seller L on Platform B, leading to a decline in product valuation and a decrease in consumer surplus.

Corollary 5. *(a) When* $\alpha_1 < \alpha < \alpha_2$ *and* $\beta > 0.95$*, we have* $\pi_A^{(A,B)*} > \pi_A^{(B,A)*}$*, and* $\pi_B^{(A,B)*} > \pi_B^{(B,A)*}$ *if and only if* $\beta < \beta_6$ *or* $\alpha > 0.43$*; (b) when* $\alpha\beta_3 < \beta < \beta_3$ *and* $\beta(1-\alpha)(\beta-\alpha)(4-\beta)^2 < \left((2\beta+1)\alpha+\beta^2-4\beta\right)^2\alpha$*, we have* $\pi_B^{(AB,B)*} > \pi_B^{(A,A)*}$*, and* $\pi_A^{(AB,B)*} > \pi_A^{(A,A)*}$ *if and only if* $\alpha < \alpha_3$*.*

Note: the expressions of α_3 *and* β_6 *are given in the Appendix.*

Corollary 5 compares the profits of the platforms under different sequences of seller movements. When the credit difference between the two sellers is minimal, it is more advantageous for Platform A if Seller H moves first rather than Seller L, as this scenario allows the higher-credit product to be sold on Platform A. For Platform B, Seller H moving first is beneficial when the credit difference between platforms is small; in this case, the advantage of reduced competition between the sellers outweighs the negative impact of a lower-credit product being sold on Platform B. Furthermore, when the credit differences between the sellers and platforms are both moderate, Platform B prefers Seller H to move first; otherwise, it risks not being selected by either seller. Interestingly, Platform A may be worse off if Seller H moves first. If the credit difference between platforms is small, the benefit of monopolizing the market for Platform A is overshadowed by the loss incurred from intensified competition between the two sellers. In this case, Platform A would prefer Seller L to move first or for the sellers to move simultaneously, rather than having Seller H move first.

We conclude this section with a discussion on the impact of the "either-or" policy. Based on the previous analysis, we find that the equilibrium homing strategy remains unchanged under the "either–or" strategy when the sellers move simultaneously or Seller L moves first. The only change caused by the "either-or" policy is that the equilibrium (AB, B) when Seller H moves first will transform into (A, A) if the credit differences between the sellers and platforms are both in the medium range, which expels Platform B and makes Platform A a monopoly-selling channel. However, as shown in Corollary 5 (b), this does not necessarily benefit Platform A. When the credit of Platform B is relatively low, the "either-or" policy leads to a lose-lose situation for both platforms, as shown in Fig. 8. Considering the results of Corollary 2, the purpose of promulgating this policy would have been to circumvent the losses caused by the multi-homing behaviour of the high-credit seller, but this goal can only be achieved if the platform selection of the low-credit seller remains unchanged. In this scenario, as Seller H switches from multi-homing to single-homing to Platform A, Seller L will also migrate from Platform B to Platform A, resulting in increased competition that can be to the detriment of Platform A. For the sellers, it is straightforward that the "either-or" policy is unfavourable to Seller H and favourable to Seller L. In addition, the "either-or" policy would disadvantage Seller H while benefiting Seller L, as requiring Seller H to adopt single-homing would eliminate his first-mover advantage. Finally, as shown by Corollary 4(b), this policy can increase total consumer surplus since the more intense competition resulting from equilibrium transformation leads to lower prices and higher demands.

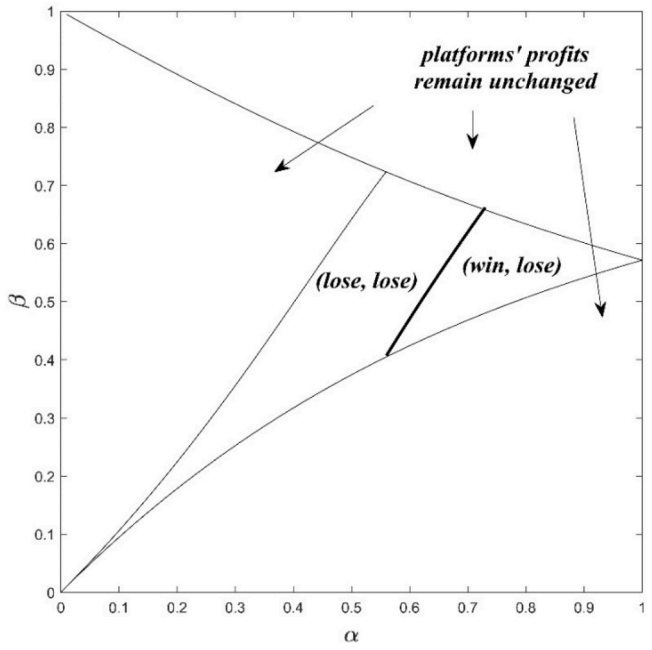


Fig. 8. The effects of the "either-or" policy on platforms ($\lambda = 0.1$).

6. Extensions

6.1. Different platform commission rates

In this section, we extend the model to the case of different platform commission rates. Suppose that Platform A and Platform B charge percentage commission rates λ_A and λ_B, respectively, with $\lambda_A > \lambda_B$. All the solutions of the nine strategy combinations are shown in the Appendix, and the equilibrium homing strategies corresponding to the three types of moving sequences are illustrated in Fig. 9. A major difference from the main model is that Platform B is adopted more by the sellers due to a commission rate lower than that of Platform A. This transforms the original region (A, A) into (A, AB) and (AB, AB) as two additional multi-homing strategy equilibria when the credit of Seller L is lower than certain thresholds.

Despite this change, the major insights obtained in the main model carry over to the case of different platform commission rates. That is, when the credit difference between the sellers is small, single-homing strategies are still adopted by the sellers to avoid competition intensification. Platform A is still the first choice for Seller H in most situations, while Platform B can still be adopted by Seller L to hedge the competition, although it lowers consumers' WTP. In addition, the homing strategy choice is a first-mover advantage game for the sellers when the credit difference between the sellers is significantly small and when the credit differences between the sellers and platforms are both in the medium range; in other cases, the equilibrium homing strategy outcome is not relevant to who enters the market first. Furthermore, the "either-or" policy can still result in lose-lose for the platforms, although in this case the side effects for sellers are more prominent.

6.2. Endogenous platform commission rates

In this section, we consider endogenous platform commission rates, which are determined by the two platforms to maximize their respective profits before the sellers make their homing and pricing decisions. We conduct a numerical study to analyse the equilibrium homing strategy, as illustrated in Fig. 10. Similar to the findings in Section 6.1, in this scenario, Platform L sets a lower commission rate than Platform H, thereby enhancing the incentives for sellers to adopt a multi-homing strategy. Other than this, the key insights derived from the base model are still applicable. Specifically, when credit competition between the two sellers is intense, both sellers are likely to adopt single-homing strategies to mitigate competition. Furthermore, if one seller acts first, he still has a first-mover advantage by compelling the other seller to choose the inferior platform. Additionally, we also observe that the "either-or" policy can lead to a lose-lose situation for the platforms when the credit differences between sellers and between platforms are both within a medium range.

6.3. Two-Dimensional vertical differentiation model

In this subsection, we consider two independent variables s and v to represent consumers' valuations for Platform A's credit and Seller H's credit, respectively. The corresponding discount factors are $\alpha \in (0,1)$ and $\beta \in (0,1)$ for Platform B and Seller L, respectively. Thus, a customer

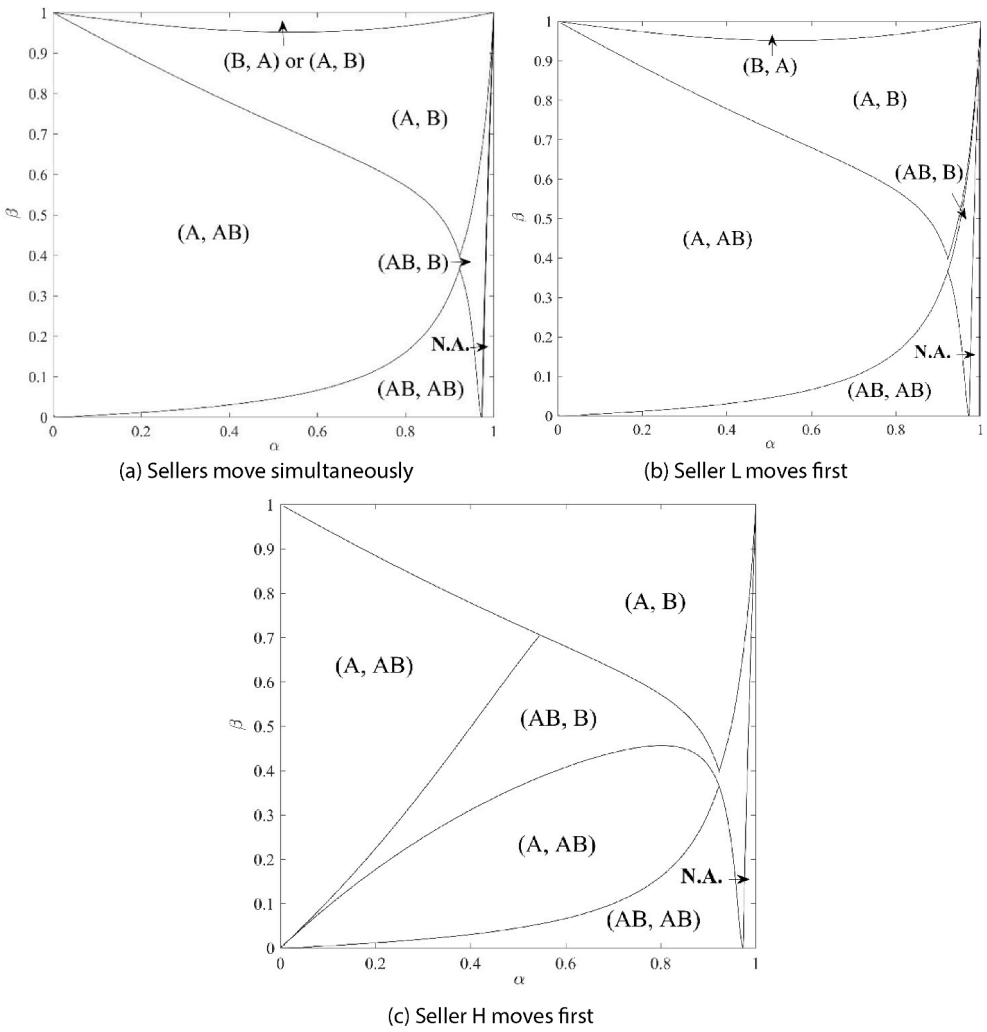


Fig. 9. Equilibrium homing strategy combination ($\lambda_A = 0.1$, $\lambda_B = 0.05$).

will obtain a net utility: (a) $u_{HA} = s + v - p_{HA}$ if purchasing from Seller H on Platform A; (b) $u_{HB} = \alpha s + v - p_{HB}$ if purchasing from Seller H on Platform B; (c) $u_{LA} = s + \beta v - p_{LA}$ if purchasing from Seller L on Platform A; and (d) $u_{LB} = \alpha s + \beta v - p_{LB}$ if purchasing from Seller L on Platform B. We suppose that s and v are independent and uniformly distributed over [0,1] among consumers.

Under this consumer willingness-to-pay model setting, we present the homing strategies corresponding to the three types of moving sequences illustrated in Fig. 11. Our key insights from the main context remain valid. For instance, sellers with little credit difference prefer to join different platforms to reduce competition, and a first-move advantage is manifested in this case. As the credit difference between sellers increases, joining Platform A becomes the dominant strategy for Seller H, while Seller L's choice depends on his relative credit if the sellers move simultaneously or if Seller L moves first. Moreover, when the credit difference between sellers and that between platforms fall within certain medium ranges, Seller H will adopt a multi-homing strategy if she moves first. In this case, the implementation of "either-or" policy can cause a lose-lose situation for both platforms.

6.4. Platform reselling mode

In the previous analysis, we assume that platforms take a percentage commission from the sellers' revenue, a model commonly referred to as the agency selling mode. In this subsection, we consider the reselling mode, where sellers sell to the platform under a wholesale price contract and the platform subsequently resells to consumers. In practice, most e-commerce platforms, such as Tmall, Taobao, Pinduoduo, TikTok Shop, and WeChat Store, primarily utilize the agency selling mode, while JD and Amazon employ both agency and reselling modes. In the reselling mode, we consider that Seller i sells the product to Platform j at a wholesale price w_{ij}, and then Platform j sells them to consumers at a retail price $w_{ij} + m_j$, where m_j denotes the profit margin of Platform j. We suppose that w_{ij} is the decision variable for Seller i while m_j is exogenous.

The optimal solutions are presented in the Appendix, and the sellers' equilibrium homing strategies are illustrated in Fig. 12. It is shown that similar to the results in the base model, the sellers are likely to join an inferior platform when competition among them is radical. As the credit difference between sellers increases, Seller H prefers to join Platform A, while Seller L may choose either Platform B or multi-homing. Additionally, the result that the seller who moves first can compel her competitor to the inferior platform remains unchanged. On the other hand, there are also different results under the reselling mode. First, (A, A) represents the equilibrium homing strategies when the credit difference between platforms is significant and the difference between sellers is not too small. In this case, the double marginalization effect results in a selling price too high for consumers to purchase on Platform B, and thus (A, A) becomes a unique equilibrium. Second, (AB, B) serves as an equilibrium when the credit difference between platforms is minimal while the difference between sellers is substantial. In this case, Seller H chooses to expand her market share by multi-homing, while Seller L opts to sell on Platform B to mitigate competition.

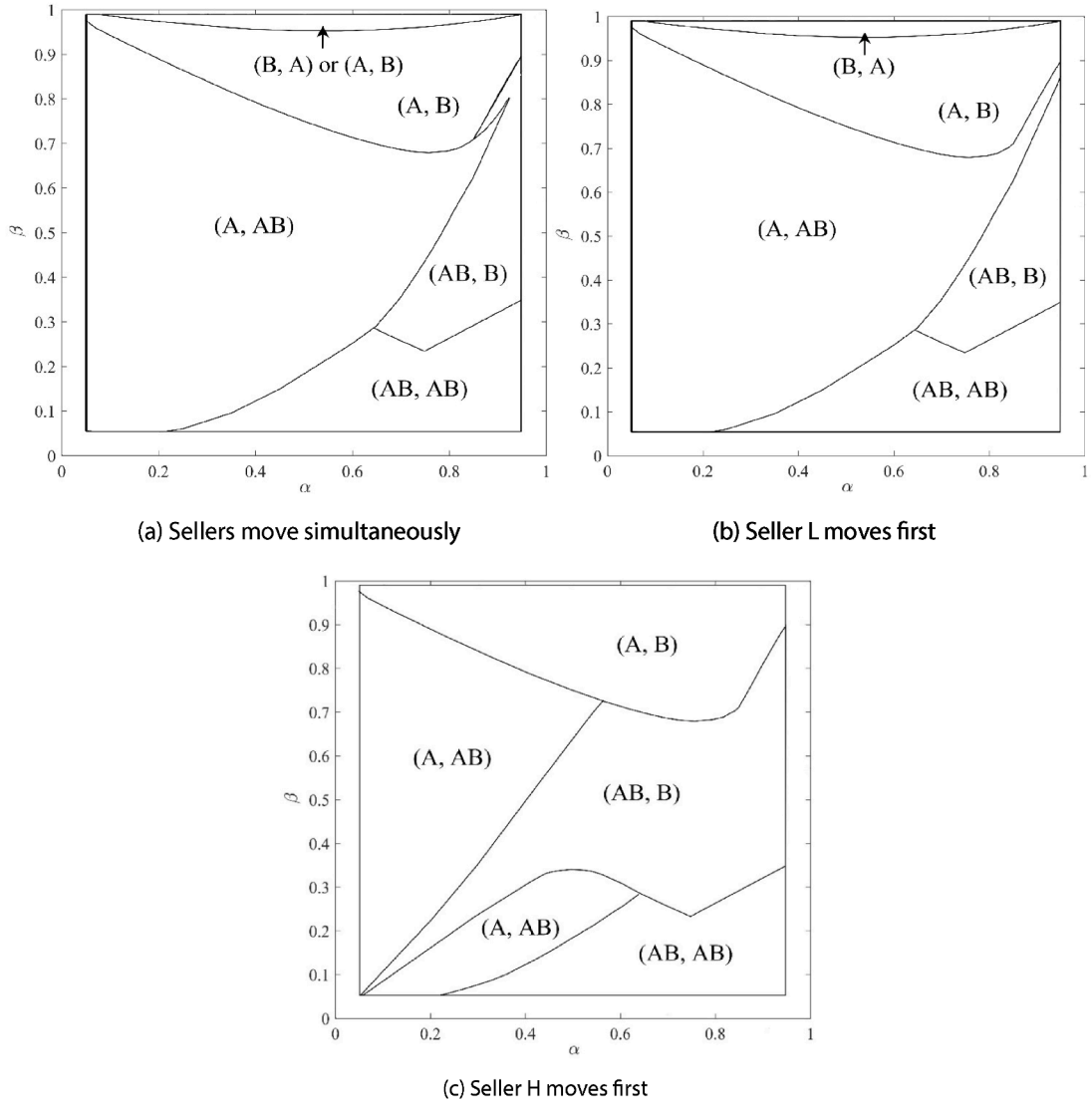


Fig. 10. Equilibrium homing strategy combination with endogenous platform commission rates.

6.5. Fix costs for expanding channels

In this section, we assume that a fix cost F will be incurred if a seller operates on an additional platform as selling channel. In this scenario, when the sellers adopt a multi-homing strategy, their profit functions are formulated as $\pi_H^{(AB,n)} = (1-\lambda)\left(p_{HA}^{(AB,n)} D_{HA}^{(AB,n)} + p_{HB}^{(AB,n)} D_{HB}^{(AB,n)}\right) - F$ and $\pi_L^{(m,AB)} = (1-\lambda)\left(p_{LA}^{(m,AB)} D_{LA}^{(m,AB)} + p_{LB}^{(m,AB)} D_{LB}^{(m,AB)}\right) - F$, respectively. As a result, the equilibrium homing strategies remain unchanged in the presence of channel expansion costs if Seller L moves first or if the sellers move simultaneously. However, if Seller H moves first, the costs will diminish Seller H's inclination to adopt a multi-homing strategy when the credit gaps between sellers and platforms are moderate, as illustrated in Fig. 13. In other words, the first-mover advantage of Seller H is weakened, which disadvantages Seller H while benefiting Seller L. In addition, Platform B is worse off as these costs lead to a greater tendency for the equilibrium strategy combination (A, A). Meanwhile, Platform A is also adversely affected when the credit of Platform B falls below some threshold (i.e., $\alpha < \alpha_3$), in which case she benefits from the multi-homing behaviour of Seller H. Furthermore, total consumer surplus increases since the strategy combination (A, A) leads to lower prices and higher demands. Finally, the impact of "either-or" policy becomes smaller because the costs of expanding channels inherently restrict sellers from implementing a multi-homing strategy.

7. Conclusion

In this paper, we investigate the homing strategies of asymmetric sellers on differentiated platforms. A key element of our research is that the two sellers and platforms differ from each other in terms of their credits to consumers. We use a stylized consumer willingness-to-pay model to describe this characteristic. On the sellers' competition side, we develop a two-stage game where the sellers first choose their platform-homing strategies and then determine the selling prices on the selected platform(s). This analytical framework allows us to examine how the discrepancies between sellers and platforms influence sellers' homing strategies.

7.1. Major findings

The main results found by this paper are summarized as follows: First, the value of the inferior platform should be highlighted as a buffer in the competition among sellers. When there is a moderate difference between the sellers, the seller with lower credit will choose to join the inferior platform voluntarily, even if it may decrease the perceived value of their products among consumers. In this case, this valuation-reducing

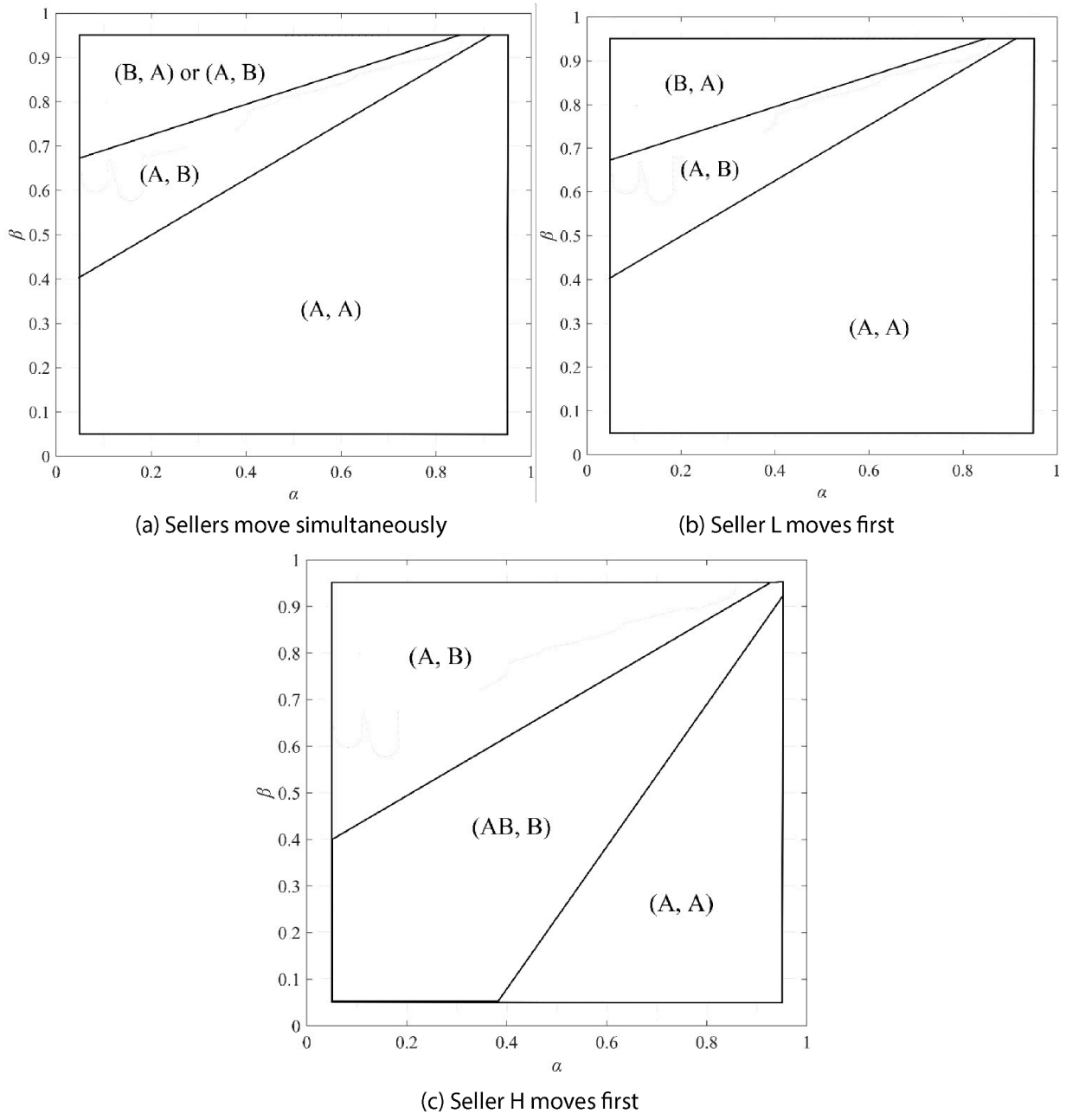


Fig. 11. Equilibrium homing strategy combination with two-dimensional vertical differentiation ($\lambda = 0.1$).

effect is offset by the charged higher selling price due to competition mitigation, the latter referred to as the competition-mitigating effect. Similarly, if the seller with higher credit finds that the superior platform is already taken by her competitor, she will also opt for the inferior platform when the credit difference between them is minimal.

Second, the negative impact of multi-homing should be emphasized compared to single-homing. Specifically, single-homing is not a special case of multi-homing but rather a strategy that may be strictly superior to multi-homing: in the context of intense competition among sellers, multi-homing will lead to the competition-intensifying effect, and the negative impacts it brings outweigh the benefits of the market-expanding effect resulting from the opening of new sales channels. Consequently, multi-homing can be detrimental to sellers, even if there are no additional costs associated with opening the new marketplace. For instance, under a small disparity in credits between two sellers, both parties adopt a single-channel strategy that leads to equilibrium.

Third, when the gap between sellers is small, the competition between them become so intense that multi-homing is a dominated strategy for both sellers, and a first-mover advantage is manifested in the single-homing strategy selection. When the credit gaps between sellers and platforms are in some medium ranges, the superior seller will adopt a multi-homing strategy when she moves first, causing a different equilibrium outcome than when both sellers act simultaneously or when it is the inferior seller that moves first, also creating a game of first-mover advantage. In other cases, the equilibrium of the platform choice game can be irrelevant to the order of actions, and thus the first mover does not always benefit.

Finally, the original intention of the “either-or” policy was to restrict sellers, especially high-credit sellers, from multi-homing to increase revenue for the platform. However, it is based on the false assumption that the choice of the low-credit seller does not change with the homing strategy of the high-credit seller. In fact, considering the interaction of sellers’ competitive strategies, the “either-or” policy not only disadvantages the inferior platform by causing traffic losses but also may

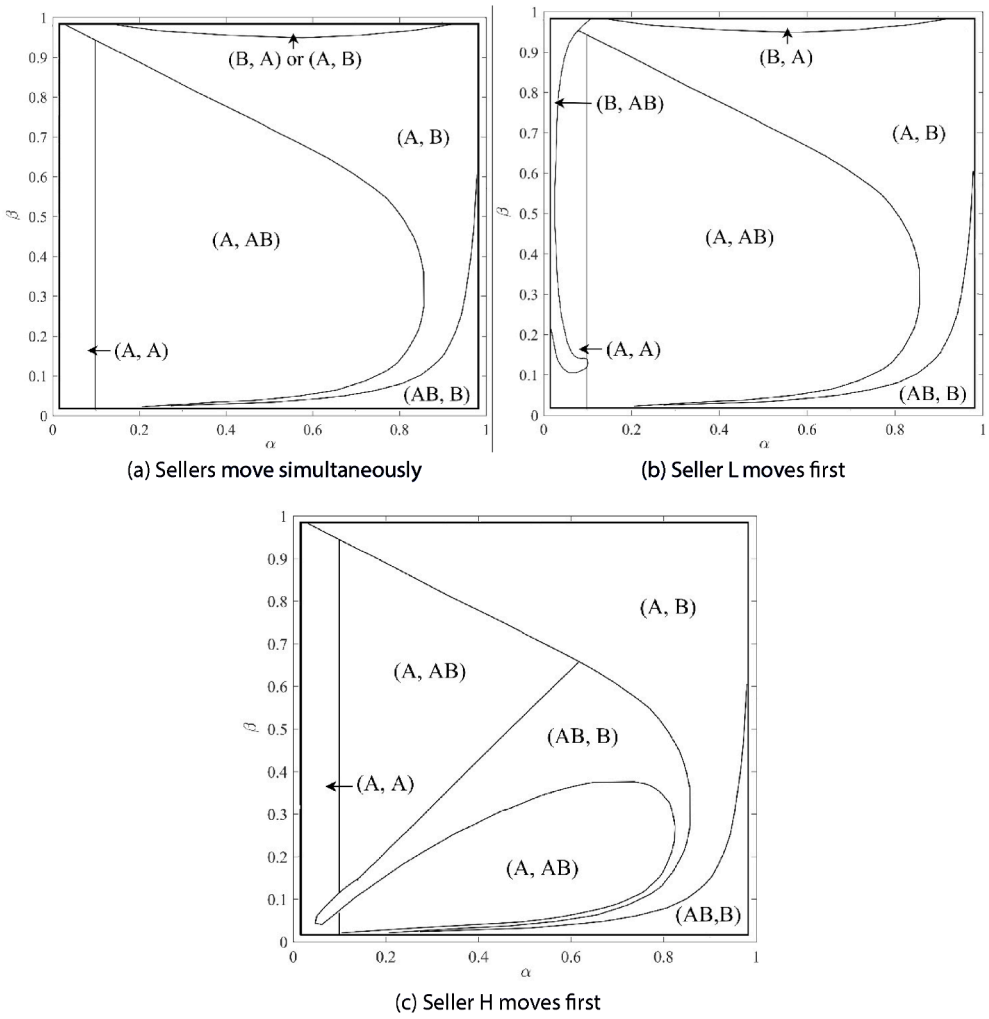


Fig. 12. Equilibrium homing strategy combination with reselling contract ($m_A = 0.01, m_B = 0.001$).

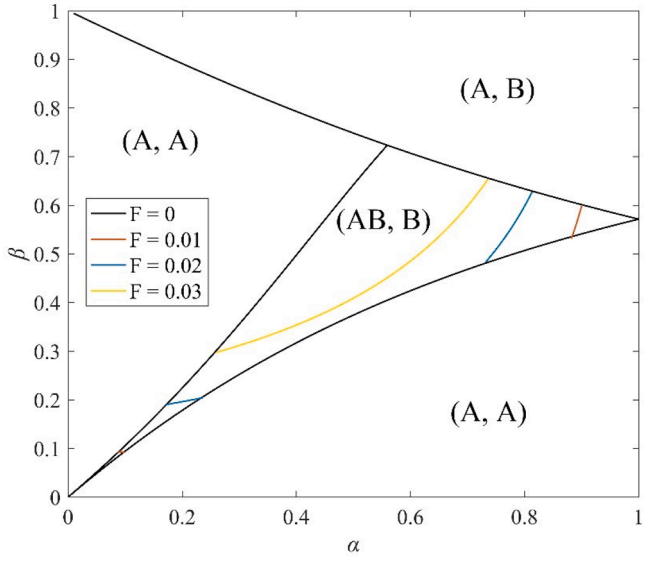


Fig. 13. Equilibrium homing strategy combination when Seller H moves first ($\lambda = 0.1$).

harm the superior platform, thus creating a lose-lose situation, especially when there is a significant gap in platform credit. Furthermore, contrary to popular belief, imposing the "either-or" policy can be beneficial for both the low-credit seller and the consumers.

7.2. Managerial insights

Our analytical findings provide valuable managerial implications for practicing managers. When confronted with the problems of platform homing and pricing strategies, practitioners should carefully assess various business environment factors, including the intensity of seller competition, differences in platform credits, and the strategic behaviors of rivals. First, it is unnecessary for sellers to be obsessed with joining high-credit platforms. In fact, joining platforms of relatively lower credit can be beneficial to both high-end and low-end sellers by mitigating seller competition and raising selling prices. Secondly, it is not always advantageous for sellers to join multiple platforms. In the context of highly competitive sellers, single-homing is a better strategy than multi-homing, as the competition-mitigating effect of single-homing can overweigh the market-expanding effect of multi-homing. Thirdly, sellers do not always enjoy a first-mover advantage when selecting platforms. Sellers should be more proactive in establishing their presence on platforms, such as launching on a high-credit platform or adopting a multi-homing strategy, in more competitive markets. However, this prioritization of platform occupancy may yield limited benefits when there are significant credit disparities between sellers or platforms. Finally, the

dominant platform should be prudent when considering the implementation of an "either-or" policy. When there is a significant disparity in the platforms' credits, the "either-or" policy may not only restrict competitors but also harm its own profit.

7.3. Future directions

Like any other research, our work has limitations. First, we only focus on two sellers as representatives of the vast number of sellers. Future research could consider scenarios in which multiple sellers have different credit levels or even continuous types of sellers competing on platforms. Second, we consider the credit of the sellers and platforms to be exogenous. In reality, however, a firm's reputation can change over time, and attributes such as product quality and service levels can improve in a shorter period. Therefore, it is important to further investigate situations where credits can be enhanced by exerting corresponding efforts. Finally, our research focus on the e-commerce platforms and sellers with physical products. Other platforms, such as on-demand service platforms and their service providers, can encounter similar problems. However, these platforms have distinguished features including supply and demand matching, service capacity allocation, and cross-side network externalities. Although these issues are beyond the scope of this paper, conducting further research that takes these factors into account represents an important avenue for future studies.

CRediT authorship contribution statement

Xiang Li: Writing – review & editing, Supervision, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization. **Meiqi Wang:** Writing – original draft, Software, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this article.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.omega.2025.103284.

Data availability

No data was used for the research described in the article.

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